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Metaphorical Framing of Climate Change in Chinese and American News Media: A Corpus-assisted Discourse Study

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ABSTRACT

Climate change is both a scientific and political issue. Addressing this issue adequately requires not just scientific advancement but also an enhancement of social cognition. China and the United States, as two major global powers, play pivotal roles in shaping international climate policies. Their media outlets provide distinct lenses through which climate change is conceptualised. However, few studies have compared metaphorical framing of climate change in newspapers of the two countries. This study marries framing theory with the analysis of conceptual metaphors to investigate how climate change is conceptualised in two newspapers: *China Daily* (CD), China's official English-language newspaper, and the *New York Times* (NYT), a leading US newspaper that shapes the national media agenda. Findings reveal CD's preference for WAR, JOURNEY, MORALITY and BUILDING metaphors to emphasise the need to address climate change. It tends to construct the Chinese government as proactive in tackling the issue. By contrast, NYT uses LIE, BUSINESS, DISASTER, and JOURNEY metaphors to emphasise uncertainty, economic concerns, and policy obstacles. These specific metaphorical choices indicate NYT's critical stance on current climate policies, but may reinforce neoliberal ideology. These findings are expected to offer valuable insights for strengthening governance efforts to address climate change.

KEYWORDS

Climate change; conceptual metaphor; framing; corpus-assisted discourse study; critical discourse analysis

Introduction

As the world's two largest carbon emitters, China and the US significantly impact global climate governance and shape international agendas (Fu and Wang 2022). However, their different national interests and dominant ideologies lead to different stances on climate change. As a leading developing country, China has increasingly embraced a proactive role in global climate governance (Song et al. 2021). It adopts a top-down model and advances substantial domestic and international climate initiatives (Wang, Cui, and Qu 2024). In contrast, policy directions in the US fluctuate with political administrations and the priorities they assign (Gong, Liu, and Sun 2020; Selin and VanDeveer 2020).

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Climate policies generally operate within a neoliberal framework (McGee 2014). Democratic administrations prioritise climate action as a pathway to economic growth, while Republican administrations view it as restrictive to market practices. In this context, US newspapers play a critical role in scrutinising governmental climate actions, especially concerning environmental policies (Hellmueller and Mellado 2016).

News media are pivotal in shaping public understanding of climate change through defining issues, attributing causes and impacts, and proposing solutions in particular ways. Their highlighting of certain aspects while obscuring others (Entman 1993) influences public and governmental decisions on climate actions (Pan, Opgenhaffen, and Van Gorp 2019). In turn, news framing of climate change is greatly affected by media systems, national interests and dominant ideologies, leading to significant differences in how climate change is portrayed across countries (Liu and Huang 2024; Song et al. 2021). Chinese media often portray climate change as a global challenge (Song et al. 2021), emphasising solutions and cooperation (Fu and Wang 2022), while US media tend to frame it as a local risk (Su and Hu 2021), highlighting crisis and polarisation (Parks 2020; Wang and Habil 2024). Additionally, Chinese newspapers underline the certainty of climate change, whereas US newspapers tend to highlight controversies surrounding climate change, its human causes and its impacts (Liu and Huang 2024). These framing patterns indicate that Chinese media emphasise their government's commitment to global responsibilities, while US media focus more on its government's reluctance toward strong climate actions. Most related studies rely on content analysis, incorporating quantitative and qualitative analyses to categorise texts based on various themes to identify patterns in media communication about climate change. While an increasing number of studies have explored the cognitive underpinnings behind representations of climate change in Chinese and American newspapers (e.g. Wang and Habil 2024), few have addressed the political interests shaping media narratives.

Therefore, this study intends to conduct a critical metaphor analysis on the media framing of climate change in two leading English-language newspapers in China and the US, *China Daily* (CD) and *the New York Times* (NYT) to examine socio-political factors that influence media representations. Since the introduction of conceptual metaphor theory (CMT) by Lakoff and Johnson (2008), metaphor has been understood as a matter of both language and thought (Liu 2015). Conceptual metaphors play a crucial role in shaping how people understand abstract climate topics by conceptualising climate change through familiar knowledge, thus influencing how people think about solutions, responsibility, and urgency in subtle ways (Koteyko and Atanasova 2017). Studying conceptual metaphors in news discourse about climate change can therefore provide illuminating insights into how language shapes public perception and attitudes. Furthermore, this approach can uncover implicit political motivations and ideologies (Atanasova and Koteyko 2017a; Cohen 2011). By highlighting certain aspects of reality while hiding others, conceptual metaphors can legitimize specific climate policies and advance the interests of particular groups (Atanasova and Koteyko 2017a). CD and NYT were chosen because they are key players in expressing their respective countries' national voices and protecting their national interests in the international arena (Lee et al. 2002). CD is noted for its important role as China's official English-language newspaper, expressing the Chinese government's political viewpoints to the world (Liu and Huang 2024). NYT, known as a "newspaper of record" in the US, significantly shapes the national media agenda, sets

standards for other newspapers, and influences the content of various mass media (Liu and Huang 2022). This study aims to answer the following research questions: (1) How is climate change framed through conceptual metaphors in CD and NYT? and (2) What are the hidden stances of CD and NYT toward climate change? The findings are further discussed in relation to media systems, national interests and dominant ideologies, illuminating the relations between conceptual metaphors and broader socio-political factors.

Conceptual metaphors and climate change

Conceptual metaphors are effective in climate change communication by framing abstract ideas in terms of familiar concepts, making complex climate issues more accessible (Asplund 2011). Through connecting two domains, metaphors shape public perceptions, emphasising specific aspects while downplaying others (Atanasova and Koteyko 2017a). Common metaphors include war (Flusberg, Matlock, and Thibodeau 2017), race/game (Asplund 2011), tipping points (Russill and Nyssa 2009), greenhouse effect (Bostrom 2008), health/illness (Atanasova and Koteyko 2017a), machines (Luokkanen, Huttunen, and Hildén 2014) and religion (Woods, Fernández, and Coen 2012). These metaphorical choices help convey complex concepts regarding climate events, impacts, and responses. For example, Russill and Nyssa (2009) found that the metaphor of tipping points underscored the dangers of climate change and evoked public anxiety. Likewise, Nerlich and Hellsten (2014) suggested that the metaphor of greenhouse effect served as a scientific basis for framing climate change as a threat.

Conceptual metaphors can serve as powerful political tools to influence people's opinions and legitimize political agendas (Atanasova and Koteyko 2017a). Metaphorical framing in the media is deeply related to socio-political factors, such as partisan orientation and ideologies. For example, Atanasova and Koteyko (2017a) discovered that *Guardian Online*, the left-leaning UK newspaper, frequently used war metaphors to stress climate urgency, whereas *Mail Online*, the right-leaning newspaper, favoured religion metaphors to downplay urgency and foreground climate skepticism. Similarly, Nerlich and Koteyko (2010) observed that UK financial newspapers used gold rush metaphors to highlight financial opportunities of carbon mitigation schemes, while general newspapers were more likely to use cowboy imagery to underline the risks of exploitation. However, most studies have focused on Western media, particularly in the UK and the US (e.g. Atanasova 2022; Atanasova and Koteyko 2017b; Hellsten, Porter, and Nerlich 2014). Comparative analyses of the metaphorical framing between Chinese and American newspapers remain limited (exceptions such as Wang and Hadina 2024; Zhang and Gai 2022). For instance, Zhang and Gai (2022) compared conceptual metaphors in Chinese and American news discourse in 2017–2019 from the ecolinguistic perspective. They identified different ecological orientations behind different metaphors, finding that Chinese newspapers tended to follow the ecological view of "harmonious coexistence between man and nature", while US newspapers highlighted the ecological view of "America First". However, the existing literature focused on a relatively short time frame and gave limited attention to the political motivations and national interests influencing these metaphors. Given the crucial roles of China and the US in addressing climate change since the 2000s, it is important to further explore how conceptual

metaphors in their media are used to frame climate change and how these choices relate to broader socio-political factors.

Critical metaphor analysis

To effectively compare the metaphorical framing of climate change in Chinese and American newspapers, this study draws on the theoretical framework of Critical Metaphor Analysis (CMA) proposed by Charteris-Black (2004). CMA emphasises the role of metaphor analysis in uncovering viewpoints on specific issues and revealing underlying political and ideological motivations. It integrates critical discourse analysis (CDA), conceptual metaphor theory (CMT), pragmatics, and corpus linguistics (CL), highlighting that conceptual metaphor involves cognitive, linguistic, and pragmatic dimensions. Metaphor is cognitive because it can shape the ways in which people mentally structure complex issues. It is pragmatic in that it serves not only as a rhetorical tool to capture audience's attention but, more importantly, as a linguistic tool to convey evaluations of certain issues and ultimately shape beliefs, attitudes and actions. Charteris-Black (2004) outlines three stages of CMA: identification, interpretation, and explanation. Identification involves recognising metaphorical expressions in the text. Interpretation reveals the cognitive structures underlying the metaphor. Explanation examines the broader socio-political and ideological implications of metaphors. By underlining "salient aspects of a socially, culturally, or politically relevant topic" (Atanasova and Koteyko 2017a, 454), conceptual metaphor can frame an issue in particular ways, communicate ideological meanings and justify particular political actions. The analysis of conceptual metaphor is thus important in CDA, which views discourse as a social practice and seeks to explicate relations between discourse and ideology.

CMA also highlights the potential to employ approaches from CL and integrate quantitative calculation with qualitative interpretation in metaphor analysis. The combination can not only help identify subtle metaphorical patterns that are difficult to detect through frequency and typicality but also facilitate more effective and objective data processing. Additionally, it enables a deeper understanding of the pragmatic role of metaphors, revealing underlying stances and ideologies on specific issues (Liu 2015).

Given that CMA approach can effectively dissect the framing of climate issues and analyse the interplay between language and ideology in climate change narratives, this study employs this approach to examine how conceptual metaphors frame climate change and convey political interests and ideologies in two leading newspapers from China and the US.

Data and methodology

This study analyses news reports from CD and NYT. Two corpora were built, namely the CD corpus and the NYT corpus. To investigate the metaphorical framing of climate change over two decades, reports with "climate change" in the headline from January 1, 2001 to December 31, 2020 were collected from online databases Factiva (CD) and LexisNexis (NYT). The period from 2001 to 2020 was selected to capture key developments in climate discourse. The starting point, 2001, marked increased media attention following the IPCC Third Assessment Report, while the ending point, 2020, was chosen to minimise

the impacts of the COVID-19 pandemic on news coverage (Stoddart et al. 2021). This timeframe included events like the US' withdrawal from the Kyoto Protocol and the Paris Agreement. To make sure that all the collected data were relevant to the topic, the downloaded reports were further manually checked. Reports that mentioned climate change only briefly, such as "Coronavirus, U.K. Quarantine, Climate Change: Your Monday Briefing" in 2020, were excluded from the analysis. The final sample comprised 438 news reports in the CD corpus (291,958 tokens) and 1,071 reports in the NYT corpus (1,106,660 tokens). The lower volume in CD reflects comparatively less focus on climate reporting, which may be attributed to the Chinese government's limited attention to climate issues before 2007 (Broadbent et al. 2016). Figure 1 illustrates the diachronic distribution of news reports in CD and NYT. It reveals that coverage of climate change in CD has remained relatively stable over time, while NYT has witnessed a significant surge in climate reports since 2014.

This study follows Charteris-Black's (2004) three-stage framework: metaphor identification, interpretation, and explanation. The first stage was to identify metaphor keywords related to climate change in the news reports. Concordance lines of the keywords "climate change" were extracted to identify conceptual metaphors relevant to climate change. Metaphor keywords included content words, such as nouns, verbs, adjectives, adverbs, and some prepositions, such as *up*, *away*, *back*, etc. Some prepositions were included as they often described the progression or regression of climate actions. Function words, such as auxiliary verbs, conjunctions, determiners, etc., and delexical verbs, such as *do*, *get*, *have*, etc., were excluded (Yu 2019). Whether the expressions were used metaphorically depended on whether they had a more concrete meaning in other contexts (Pragglejaz Group 2007). Every concordance line was examined and candidate keywords were selected. Collins Dictionary was used as a reference to decide whether they were used metaphorically. The second stage concerned metaphor interpretation, which involved the identification of "conceptual metaphors" or "conceptual keys" (Charteris-Black 2004, 38). This process specifically concerned how conceptual metaphors were used to frame different aspects of climate change. The metaphor keywords were categorised into different framing groupings based on their semantic meanings in their literal senses, and their frequencies in a particular corpus were examined. Some groupings were also classified into different "metaphor scenarios" that were important to

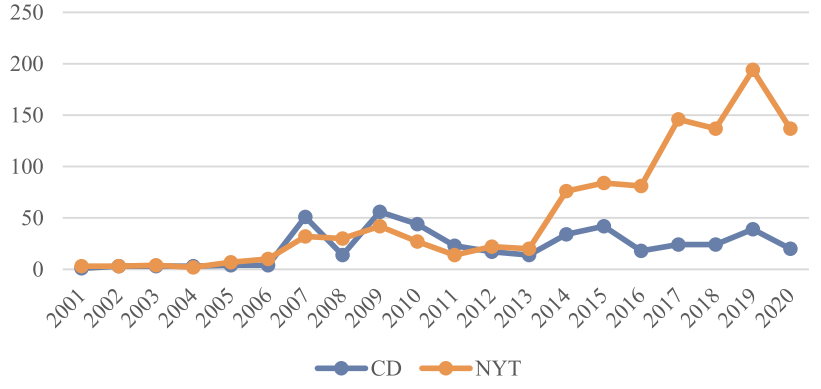


Figure 1. Diachronic distribution of news reports in CD and NYT.

construct a larger and coherent metaphorical story (Musolff 2004). Finally, the explanation step investigated the relations between metaphorical framing and wider socio-political factors. A critical examination was made to uncover the ideological implications in CD and NYT.

Findings

It was found that conceptual metaphors were mainly used to frame three aspects of climate change in CD and NYT, namely (1) climate problem; (2) impacts; and (3) solutions (Entman 1993). Among them, one type of metaphor (i.e. LIE metaphor) was discovered in framing climate problem. Two types of metaphors (i.e. DISASTER and BUSINESS metaphors) were identified in framing impacts. Five types of metaphors (i.e. WAR, JOURNEY, BUILDING, PERFORMANCE and MORALITY metaphors) were found in framing solutions. Examples are as follows:

1. Climate Problem:

Type 1: LIE metaphor

(1) President-elect Donald J. Trump has called climate change a **hoax**, and he has declared he will try to reverse the Obama administration's environmental efforts on matters … .

(NYT, December 19, 2016)

2. Impacts:

Type 1: DISASTER metaphor

(2) This is how Michelle Bachelet, United Nations high commissioner for human rights, described the **"catastrophic"** impact of climate change …

(CD, November 11, 2019)

Type 2: BUSINESS metaphor

(3) Climate change **cost** the United States economy at least $240 billion per year during the past decade.

(NYT, July 28, 2019)

3. Solutions:

Type 1: WAR metaphor

(4) Despite being the wettest region in China, many coastal cities still lack the kind of infrastructure to **combat** climate change-related storms and flooding.

(CD, July 31, 2018)

Type 2: JOURNEY metaphor

(5) President Xi Jinping said on Sunday that China is willing to cooperate with other countries to tackle climate change during the **process** of implementing post-2015 development-agenda **goals**.

(CD, September 28, 2015)

Type 3: BUILDING metaphor

(6) Xie Zhenhua, China's special representative on climate change affairs, said the comprehensive, balanced and strong package of guidelines **laid a policy foundation** for implementing the Paris agreement.

(CD, December 17, 2018)

Type 4: PERFORMANCE metaphor

(7) To the contrary, my administration is committed to a leadership **role** on the issue of climate change.

(NYT, June 12, 2001)

Type 5: MORALITY metaphor

(8) The presumptive Democratic candidate, Joseph R. Biden Jr., has **pledged** to rejoin the Paris Agreement, **vowing** that the United States would "take a back seat to no one when it comes to fighting climate change."

(NYT, May 30, 2020)

Table 1 displays the frequencies of each type of conceptual metaphors in CD and NYT. Both newspapers favoured the metaphorical framing of climate solutions, with WAR and JOURNEY metaphors most frequently used to frame actions. This is consistent with previous studies indicating that these two metaphors are largely adopted in climate change discourse to emphasise the urgency and the necessity for collective efforts (e.g. Atanasova and Koteyko 2017b). However, NYT preferred WAR metaphor, suggesting a focus on the urgency of climate action, while CD gave comparatively equal treatment to both WAR and JOURNEY metaphors, highlighting the importance of long-term efforts (Milne, Kearins, and Walton 2006). Furthermore, CD also tended to employ MORALITY, BUILDING and PERFORMANCE metaphors to foreground obligations and the step-by-step processes involved in addressing climate change.

In addition, NYT showed more interest in the metaphorical framing of climate problem and impacts. It prefers to employ LIE metaphor (334) to express the uncertainty toward climate change, reflecting ongoing disputes over climate science in US media coverage (Antilla 2005; Liu and Huang 2022; Schmid-Petri et al. 2015). The frequent use of LIE metaphor in NYT may be attributed to the journalistic norm of balanced reporting, which allows for the competing points of view presented in the media. Besides, NYT also tended to use DISASTER (199) and BUSINESS (134) metaphors to foreground the calamitous climate impacts. It may be attributed to its adherence to the journalistic norm of dramatisation (Boykoff and Boykoff 2007) to attract media attention.

The following analysis focused on conceptual metaphors employed to frame different aspects of climate change in CD and NYT. However, PERFORMANCE metaphor was not analysed since both corpora showed less interest in using it. Other metaphors were investigated in both Chinese and US contexts.

Table 1. Frequencies of each type of conceptual metaphors in CD and NYT.

Frames	Types	CD Freq.	Percentage	NYT Freq.	Percentage
Climate Problem	LIE	32	1.5%	334	12.4%
	ALL	**32**	**1.5%**	**334**	**12.4%**
Impacts	DISASTER	78	3.8%	199	7.4%
	BUSINESS	48	2.3%	134	4.9%
	ALL	**126**	**6.3%**	**333**	**12.3%**
Solutions	WAR	869	41.8%	1366	50.5%
	JOURNEY	690	33.2%	603	22.3%
	BUILDING	122	5.9%	16	0.6%
	PERFORMANCE	69	3.3%	36	1.3%
	MORALITY	171	8.2%	15	0.6%
	ALL	**1921**	**92.2%**	**2036**	**75.3%**
TOTAL		**2079**	**100%**	**2703**	**100%**

Framing climate problem: LIE metaphor

Both corpora employed LIE metaphor to illustrate the uncertainty of climate change, as manifested by the keywords such as *hoax*, *deception*, *doubt*, *deny**, etc., and the keywords related to those who disapprove of the existence of climate change such as *skeptics*, *doubters* or *deniers*.

However, CD often backgrounded climate uncertainty, as seen by the expressions such as *despite*, *still*, *even though*, *even if*, *maybe*, *in a minority*, *rejected or neglected*, etc.

(9) On the contrary, the works that ***doubt*** the authenticity of climate change are either *rejected or neglected*.

(CD, January 28, 2010)

Conversely, LIE metaphor was often expressed in phrases or clauses in NYT to designate certain groups or persons, demonstrating their explicit stance on questioning climate change. In particular, skeptical views were mostly attributed to the Republicans (31) and President Trump (89). It suggests that climate skepticism is related to partisan identities. For example,

(10) Al Gore, a prominent Democrat who served as a lightning rod for *conservatives who maintain that climate change is a* ***hoax***.

(NYT, December 2, 2014)

However, NYT also showed its disalignment of President Trump and conservatives' denial of climate science. It can be seen by the words such as *mock**, *call**, *tell**, *deride** and *claim**. Their use suggests that doubts about climate change are stated personally and subjectively (Liu and Huang 2022). This helps to underline NYT's competing views of climate change, highlighting the polarisation surrounding the issue. For example,

(11) Mr. Trump *has mocked* the science of global warming as a Chinese ***hoax*** and selected a climate-change denialist to run the Environmental Protection Agency.

(NYT, January 10, 2017)

Therefore, LIE metaphor in CD served to background skeptical views of climate change, while it was primarily used in NYT to highlight climate contrarians in the US and the increasing polarisation of the issue. Despite its negative stance toward skepticism, NYT acknowledged diverse viewpoints on climate reality and allowed for negotiation around climate science, which can hinder timely actions on climate change.

Framing impacts: DISASTER and BUSINESS metaphors

DISASTER metaphor

DISASTER metaphor was commonly employed in CD and NYT to describe the severity of climate change. It can be perceived by the adjective keywords such as *calamitous*, *destructive*, *devastating*, etc., and by the noun keywords such as *catastrophe**, *damage**, *ravage**, *toll*, etc.

However, the devastating impacts were somewhat downplayed in CD. The newspaper cautioned against an excessive focus on disasters (Example 12) and highlighted actions to deal with the impacts (Example 13).

(12) ***Doom-and-gloom*** scenarios meant to shock people into action often end up *having the opposite effect*, and so it has been at times with climate change.

(CD, November 9, 2006)

(13) Scientists who wrote the reports *focused more on how to manage risks of global warming than on* ***disaster***, Painter says.

(CD, September 25, 2014)

On the contrary, NYT employed intensification strategies, i.e. the way of intensifying the force of the utterances (Reisigl and Wodak 2016), to foreground the severity of climate change, as witnessed by such expressions as *the world's most fearsome weapon of mass destruction*, *the most calamitous consequences*, *picking up speed*, *intensified*, etc. It also highlighted hardships and inadequate efforts (22, 11.1%), portraying climate change as an issue beyond people's control. This could downplay the US' responsibilities for actions. For instance,

(14) And when climate change is depicted on screen, it's often in an onslaught of fire and brimstone, an **apocalyptic** vision that *hardly leaves room for a hopeful human response*.

(NYT, October 2, 2017)

In this sense, the disaster-based discourse can "become secretly thrilling" instead of sending "a constructive message", as it distanced people from the problem and fosters a sense of despair (Ereaut and Segnit 2006, 14). Therefore, NYT's preference for DISASTER metaphor may reduce motivation to address climate change (Painter 2013).

BUSINESS metaphor

BUSINESS metaphor can be manifested by the keywords such as *cost**, *price*, *benefit**, *interest**, etc. Two main types of metaphors were identified in both corpora, including (1) CLIMATE CHANGE is a BUSINESS, and (2) CLIMATE ACTION is a BUSINESS. Through metaphorically framing climate change as a BUSINESS, it highlighted the (un)favourable impacts associated with it. Through framing climate action as a BUSINESS, it foregrounded the expense or advantages of undertaking climate initiatives.

Although CD conceptualised climate change as a cost (19) to emphasise its unfavourable impacts, it sometimes downplayed the severity by highlighting potential benefits, particularly for the wine industry. For instance,

(15) Generally speaking for the English wine industry climate change has been *a big big bonus*, it really helps us develop.

(CD, December 26, 2015)

In addition, CD tended to frame climate action as a benefit (11) or a cost (8). On one hand, it highlighted the advantages of climate actions and climate policies (Example 16). On the other hand, it understated the costs associated with actions by suggesting that the total cost can be reduced (Example 17).

(16) Overall, the ***benefits*** of implementing the Paris Agreement on climate change *far outweigh* the ***costs***.

(CD, December 14, 2020)

(17) Technological innovation and widespread dissemination are recognised by the Chinese as an effective solution to *reduce* the ***costs*** of mitigating climate change.

(CD, June 13, 2007)

On the contrary, NYT emphasised the uncertainty of the costs of climate change, as evidenced by the verbs co-occurring with *cost**, such as *assess*, *recalculate**, *calculate*, *estimate*, *anlayzed*, *measure*, *underestimate*, *misjudged* and *undervalue**. These verbs indicates complexity in calculating the costs of climate change and may reduce the perceived urgency to act. For instance,

(18) To *assess* the **costs of climate change**, including crop failures and flooding, Professor Nordhaus developed an economic model he called the Dynamic Integrated Climate-Economy model, or DICE.

(NYT, October 9, 2018)

Furthermore, NYT also tended to foreground the costs of policy actions. It emphasised the high costs of supporting less developed countries, indicating a narrative of the US' unwillingness to aid in managing climate change. For example,

(19) But a report this month by the United Nations Environmental Program estimates that **the cost to poor countries of adapting to climate change** could *rise to as high as $300 billion annually* … .

(NYT, December 10, 2014)

To sum up, BUSINESS metaphor in CD highlighted the severity of climate change and the benefits of climate actions, legitimising efforts to address climate change. In contrast, BUSINESS metaphor in NYT emphasised climate impacts and the economic risks of policies, suggesting a neoliberal framing. Focusing on costs may fuel debates on feasibility and provoke resistance, promoting a more cautious approach to climate policy.

Framing solutions: WAR, JOURNEY, BUILDING metaphors

WAR metaphor

Two types of the WAR metaphor were identified in both corpora, including (1) CLIMATE CHANGE is a WAR; and (2) CLIMATE CHANGE POLITICS is a WAR. An examination of concordance lines showed that CD and NYT preferred to frame climate change as a war, with 858 instances (98.7%) in CD and 1,289 instances (94.4%) in NYT. It indicates that both corpora highlighted the severity of climate change and the urgency to combat it. Three main metaphor scenarios were found, including (1) pre-war scenario, which focuses on the dangers of climate change, illustrated by the keywords such as *risk**, *danger**, *threat**, etc.; (2) attack scenario, related to climate actions, featuring keywords such as *attack**, *combat**, *fight**, etc.; and (3) strategy scenario, denoting plans for success, represented by keywords such as *measure**, *strategy**, etc.

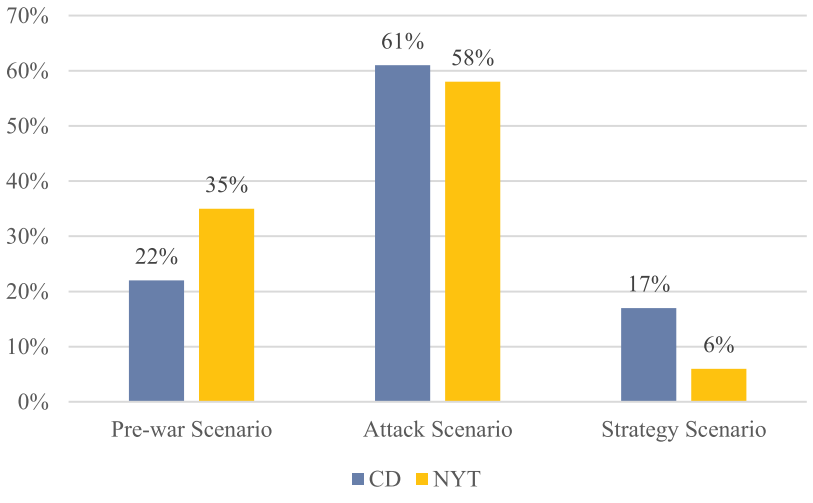


Figure 2. Distribution of the three scenarios in WAR metaphor in CD and NYT.

Figure 2 illustrates the distribution of the three scenarios in CD and NYT. It can be seen that both corpora preferred to portray attack and pre-war scenarios, conceptualising climate change as a serious issue in need of urgent actions. However, NYT showed more preference for emphasising pre-war scenario (35% compared to 22% in CD), which can potentially amplify the seriousness of climate change. It put less emphasis on strategy scenario (6% compared to 17% in CD), indicating NYT's less concern about specific actions to combat climate change.

Furthermore, NYT also highlighted a lack of confidence and failure to fight climate change (165, 12.8%). It implies uncertainty and hesitation that suggests reluctance to pursue transformative, system-disrupting actions, as in the following.

(20) **Fighting** climate change, Mr. Foglia said, is *a daunting task* that can seem impossibly broad in its scope.

(NYT, October 11, 2017)

Additionally, NYT also tended to use WAR metaphor to emphasise in-group fighting over the climate policies (77, 5.6%), shifting the narrative from "the US against climate change" to "US against THEM" (Atanasova and Koteyko 2017a). This framing emphasised division and disagreement rather than unity, reflecting the politicisation of climate change (Atanasova and Koteyko 2017a, 2017b). It may lead to a sense of paralysis and inaction, as policymakers and the public focus more on the political battles than on effective actions. For instance,

(21) Mr. Pruitt, a Republican, has been a key architect of the **legal battle against Mr. Obama's climate change policies**, actions that fit with the president-elect's comments during the campaign.

(NYT, October 7, 2016)

To sum up, WAR metaphor in CD emphasised the urgency for climate actions, whereas in NYT, it highlighted climate threats, underscored failures in combating climate change, and stressed political conflicts surrounding the issue. By highlighting failures, NYT criticised current policies and practices, calling for urgent actions. By emphasising political conflicts, NYT tended to cover diverse perspectives. However, the focus on failure and controversy may risk downplaying positive outcomes and further politicise the issue rather than foster solutions.

JOURNEY metaphor

JOURNEY metaphor was important in CD and NYT since it highlighted future goals and movements. In general, six scenarios were identified in both corpora, including (1) forward movement scenario; (2) backward movement scenario; (3) obstacle scenario; (4) direction and plan scenario; (5) path scenario; and (6) destination scenario. The forward movement scenario focuses on efforts to address climate change, evidenced by the keywords such as *advance**, *leapfrog*, *stride**, etc. The backward movement relates to a lack of progress, indicated by such keywords as *retreat*, *reverse*, *withdraw**, etc. The obstacle scenario addresses difficulties encountered along the journey, using the keywords such as *barrier**, *block*, *obstacle**, etc. The direction and plan scenario concerns the methods used to achieve progress, as shown by the keywords such as *guideline**, *direction*, *road map*, etc. The path scenario describes the route taken, indicated by the keywords such as *pathway*, *on track*, *landmark*, etc. Finally, the destination scenario refers to the ultimate goals of the journey, suggested by the keywords such as *destination*, *end point**, *goal**, etc.

Figure 3 demonstrates the distribution of different JOURNEY scenarios in the two newspapers. It shows that both CD and NYT preferred to use direction, path and movement scenarios. It aligns with Atanasova and Koteyko's (2017b) argument that the focus in JOURNEY metaphor is journeying rather than on the destination (see also Milne, Kearins, and Walton 2006).

However, CD showed more tendency to use destination scenario, highlighting its emphasis on the targets that people aim to achieve. This framing constructed the Chinese government as determined and ambitious in addressing climate change. For instance,

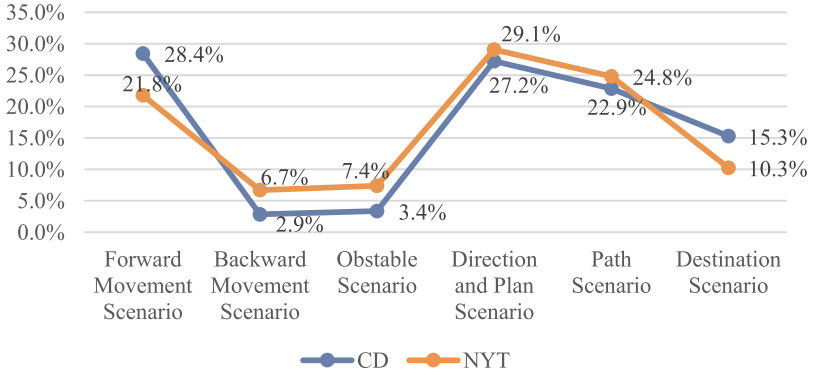


Figure 3. Distribution of six scenarios in JOURNEY metaphor in CD and NYT.

(22) **The main goal** was threefold: to launch negotiations on a climate change deal for the post-2012 period, to set the agenda for these negotiations.

(CD, December 19, 2007)

Besides, CD frequently used forward movement scenario (229, 28.4%) to emphasise progress in climate actions, positively evaluating China's efforts. This framing reinforced a narrative of success and positions the Chinese government as a leader in global climate efforts, as evidenced by the expressions such as *positive*, *encouraging*, *significant*, *great*, etc. For example,

(23) China said on Oct 4 that developing countries' right to development must be guaranteed in order to achieve a *positive* **progress** in tackling with climate change problems.

(CD, October 8, 2010)

In addition, CD backgrounded backward movement scenario (23, 2.9%) and obstacle scenario (27, 3.4%), focusing instead on hope in the climate journey. Obstacles were mostly attributed to the US (8), showing CD's unfavourable attitudes toward its inactions. For instance,

(24) Greenpeace International released a report on Nov 22, saying a handful of multinational corporations are "exerting undue influence" on the political process in *the United States and Canada* and other pivotal countries to **delay global action** on climate change.

(CD, November 26, 2011)

On the contrary, NYT foregrounded opposition to climate plans, indicating disputes and challenges in implementing climate policies. This uncertainty may help explain slower responses of the US to climate change. For example,

(25) Republican officials say the **plans** that Democrats have devised to address climate change will *decimate the economy*.

(NYT, September 3, 2019)

Besides, it put more emphasis on backward movement scenario (113, 15.8%) and obstacle scenario (53, 7.4%), which may provide an excuse for the lack of progress (Atanasova and Koteyko 2017b). The challenges in climate actions were mostly attributed to Republicans, President Trump (53), as well as fossil fuels and oil companies (11). For instance,

(26) *President-elect Donald J. Trump* has called climate change a hoax, and he has declared he will try to **reverse** the Obama administration's environmental efforts on matters …

(NYT, December 19, 2016)

To sum up, JOURNEY metaphor in CD emphasised China's long-term efforts and progress in solving climate change, portraying the country as a constructive actor. In contrast, NYT used JOURNEY metaphor to underline barriers to climate actions, reflecting a critical stance on current climate actions. While this focus underscored the challenges, it may also offer an excuse to delay progress on climate initiatives.

MORALITY metaphor

MORALITY metaphor was mostly used to emphasise the necessity of managing climate change, highlighting the ethical dimensions of the issue. It can allow for a negative evaluation of those failing to act on climate change (Atanasova and Koteyko 2017a). The metaphors were shown by the keywords such as *commit**, *pledge**, *promise**, *obligation**, *duty**, *responsibility**, etc. to underline both commitments and responsibilities.

A close examination of concordance lines revealed that CD emphasised the dedication and responsibilities of the Chinese government in tackling climate change. Positive evaluation helped to construct the Chinese government as proactive despite difficulties, as suggested by the expressions such as *make significant progress*, *firmly stick to*, *make strong commitments*, *proactive role*, *high sense of, earnestly*, etc. For instance,

(27) China *earnestly* performs its **duties** stipulated by the UNFCCC and Kyoto Protocol, released China Initial National Communications on Climate Change in 2004.
(CD, October 30, 2008)

Furthermore, CD also underlined the historical responsibilities of developed countries for offering help to less-developed countries (21). In some cases, the newspaper explicitly expressed unfavourable attitudes toward the refusal of developed countries to act on climate change (5), as in the following example.

(28) Wealthy countries are to *blame for* their failure to show international leadership on climate change and *their refusal to accept* ***responsibility*** for past carbon emissions, … .
(CD, November 28, 2007)

On the contrary, NYT underlined the commitments of the US government. It positively evaluated the climate pledges made by Presidents Obama and Biden while criticising President Trump's reluctance to make commitments. It thus implies an increasing polarisation of climate change in the US, as in the following examples.

(29) President Obama on Wednesday will ***pledge*** to *step up government aid* for Arctic communities … , *intensifying his administration's effort* to cope with the effects of climate change where they are being felt most acutely.
(NYT, September 3, 2015)

(30) President Trump signalled this year that the United States would *back away* from Obama-era ***promises*** to curb emissions.

(NYT, December 19, 2017)

Besides, NYT also tended to give a positive representation of developed countries by foregrounding their promises to reduce climate impacts, as shown in the following example.

(31) When industrialised nations ***pledged*** in 2009 to mobilise $100 billion a year by 2020 to help the poorest countries deal with climate change, it *won over some skeptics* in the developing world who had argued that rich nations should pay up for contributing so much to the problem.

(NYT, September 10, 2018)

Nevertheless, the newspaper tended to represent China as insincere in fulfilling its responsibilities regarding climate actions. It questioned China's promises and emphasised the country's desire for international recognition through its climate pledges. For instance,

(32) Foreign scientists and policy makers are also trying to judge whether Mr. Xi's 2030 ***pledge*** represents *a genuine campaign* by the Chinese government to fight climate change, or just *a business-as-usual date* when emissions would probably have levelled off anyway.

(NYT, November 13, 2014)

To sum up, in CD, MORALITY metaphor was used to emphasise China's supportive role in addressing climate change and to criticise developed countries for their inadequate actions. In contrast, NYT used this metaphor to highlight the commitments of the Obama and Biden administrations and other developed nations, while questioning China's climate pledges and condemning President Trump's failure to act urgently. These framing in CD and NYT aligned with the "ideological square" that tended to present a positive "US" against a negative "THEM" in climate actions (van Dijk 1998).

BUILDING metaphor

Climate change reduction was also conceptualised as a BUILDING in both corpora. This was evident in the keywords such as *framework*, *principle*, *basis*, *cornerstone*, etc., which foregrounded the step-by-step process in addressing climate change.

A close examination of concordance lines revealed that CD emphasised the cooperation between government and the public in dealing with climate change. For example,

(33) Through improving the *multi-ministerial decision-making coordination* mechanism and **building** an action mechanism involving a wide range of enterprise and public participation, China aims to **establish** an efficient *institutional and management framework* commensurate with the work to address climate change.

(CD, October 30, 2008)

Furthermore, CD also highlighted the necessity of following fundamental frameworks and principles in addressing climate change, such as "common but differentiated responsibilities", "the United Nations Framework Convention on Climate Change" and "Paris Agreement". For instance,

(34) He suggested the negotiations *should* stick to the basic **framework** of the United Nations Framework Convention on Climate Change (UNFCCC) and Kyoto Protocol and the mandate of the Bali Roadmap and follow the principle of "common but differentiated responsibilities".

(CD, October 8, 2010)

While NYT emphasised the foundations needed to manage climate change, it tended to focus on the difficulties of achieving consensus. This framing may risk shifting blame onto the complexity of the problem rather than recognising the need for more proactive leadership, potentially rationalising the US' failure to facilitate climate negotiation and to establish shared goals. For example,

(35) That's worth remembering as we struggle, often *despairingly*, to find common ***ground*** in the battle against climate change.

(NYT, December 7, 2019)

To sum up, BUILDING metaphor in CD was mainly used to support the establishment of common ground in international negotiations as well as to encourage nationwide and global coordination and cooperation. This framing contributed to constructing China as active and supportive in climate negotiations. However, NYT employed this metaphor to highlight challenges in achieving a global agreement, maintaining a critical stance on inaction. Nevertheless, this focus may also provide an excuse for maintaining the status-quo rather than promoting more aggressive steps.

Discussion and conclusion

The comparison of conceptual metaphors used in the CD and NYT corpora showed similarities and differences in conceptualising climate change. Both corpora favoured the metaphorical framing of climate solutions, with WAR and JOURNEY metaphors most frequently used to frame actions. However, there were key differences in their emphasis. CD tended to focus more on solutions, using metaphors like WAR, JOURNEY, MORALITY, and BUILDING to highlight coordinated actions and international commitments. It portrayed the Chinese government as proactive in combating climate change. These metaphors promoted unity and social coordination in addressing the crisis. In contrast, while NYT also employed WAR and JOURNEY metaphors, it underscored partisan conflicts and challenges. DISASTER and BUSINESS metaphors were used to highlight climate risks and policy disputes. LIE metaphor was employed to underscore ongoing contention over climate science. By repeatedly highlighting these conflicts and difficulties rather than advocating for stronger efforts, these

metaphors may risk conveying a sense of powerlessness, reinforcing the status quo within a neoliberal framework.

The conceptualisation in CD and NYT can be influenced by the media systems, national interests and dominant ideologies of the two countries. Firstly, the media systems in China and the US play a significant role in shaping climate change narratives. CD, as an official English-language newspaper, reflects a media system that is closely aligned with government policies. In this context, CD tends to frame climate change issues in line with the Chinese government's stance, emphasising proactive efforts and international cooperation. Conversely, NYT adheres to norms such as balanced reporting and dramatisation. This results in a more critical examination of climate policies, often highlighting uncertainties, conflicts, and economic implications, which can sometimes lead to informational bias that reinforces the status quo (Boykoff and Boykoff 2004). In addition, the portrayal of climate change also serves national interests of each country. For China, showcasing its commitment to addressing climate change enhances its international image and positions it as a leader in global environmental initiatives. By focusing on positive actions and international collaborations, CD reinforces China's role as a responsible major power. In contrast, NYT's emphasis on obstacles and challenges may align with the US perspective that prioritises stability and economic considerations. Highlighting the economic implications of climate regulations can serve the national interest that seeks to balance environmental action with economic growth, often through a neoliberal lens that views stringent regulations as potential barriers to economic activity. Ideological orientations further influence the metaphorical framing of climate change. In China, the prevailing ideology supports collective efforts and state-led initiatives, leading CD to stress the seriousness of climate change and the importance of coordinated action. This aligns with the government's policies that value societal harmony and collective progress. On the other hand, NYT's coverage is shaped by neoliberal ideology, which emphasises free markets and minimal state intervention. This perspective leads to framing inaction as necessary to protect the economy and portraying climate regulations as burdensome "red tape". Such framing can foster a narrative that underscores skepticism towards systemic change, thereby maintaining the status quo and reinforcing existing power structures (Chinn, Hart, and Soroka 2020; Xie 2015).

Climate change is not merely a scientific or environmental issue but is deeply rooted in values, culture, and ideology (Hoffman 2011). Analysing the metaphorical framing of climate change in Chinese and American newspapers sheds light on how political agendas and governance structures shape media discourse to align with specific policy goals and ideological perspectives. An understanding of these metaphors can help policymakers tailor their messages more effectively to different socio-political contexts, ultimately fostering greater public support for climate policies. Recognising the similarities in metaphorical framing can promote a sense of solidarity and shared responsibilities, while understanding differences can facilitate global dialogue, foster mutual respect, and ultimately contributing to more effective international cooperation and governance on climate issues. Future studies could broaden the scope by analysing post-2020 news reports and comparing other newspapers in China and the US. Additionally, examining the effects of metaphorical framing on readers could help understand how different conceptual metaphors shape public perceptions of urgency, responsibility, and potential solutions. It is expected that the study of metaphorical framing of climate change can offer

insights for improving communication and governance in addressing the global climate crisis.

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References

Antilla, L. 2005. "Climate of Scepticism: US Newspaper Coverage of the Science of Climate Change." *Global Environmental Change* 15 (4): 338–352.

Asplund, T. 2011. "Metaphors in Climate Discourse: An Analysis of Swedish Farm Magazines." *Journal of Science Communication* 10 (4): 1–8.

Atanasova, D. 2022. "How Constructive News Outlets Reported the Synergistic Effects of Climate Change and Covid-19 Through Metaphors." *Journalism Practice* 16 (2-3): 384–403.

Atanasova, D., and N. Koteyko. 2017a. "Metaphors in Guardian Online and Mail Online Opinion-Page Content on Climate Change: War, Religion, and Politics." *Environmental Communication* 11 (4): 452–469.

Atanasova, D., and N. Koteyko. 2017b. "Metaphors in Online Editorials and Op-Eds About Climate Change, 2006–2013: A Study of Germany, the United Kingdom, and the United States." In *The Role of Language in the Climate Change Debate*, edited by K. Flottum, 71–89. New York: Routledge.

Broadbent, J., J. Sonnett, I. Botetzagias, M. Carson, A. Carvalho, Y.-J. Chien, C. Edling, D. Fisher, G. Giouzepas, and R. Haluza-DeLay. 2016. "Conflicting climate change frames in a global field of media discourse." *Socius* 2 (2): 1–17.

Bostrom, A. 2008. "Lead is Like Mercury: Risk Comparisons, Analogies and Mental Models." *Journal of Risk Research* 11 (1-2): 99–117.

Boykoff, M. T., and J. M. Boykoff. 2004. "Balance as Bias: Global Warming and the US Prestige Press." *Global Environmental Change* 14 (2): 125–136.

Boykoff, M. T., and J. M. Boykoff. 2007. "Climate Change and Journalistic Norms: A Case-Study of US Mass-Media Coverage." *Geoforum* 38 (6): 1190–1204.

Charteris-Black, J. 2004. *Corpus Approaches to Critical Metaphor Analysis*. Hampshire: Springer.

Chinn, S., P. S. Hart, and S. Soroka. 2020. "Politicization and Polarization in Climate Change News Content, 1985–2017." *Science Communication* 42 (1): 112–129.

Cohen, M. J. 2011. "Is the UK Preparing for "War"? Military Metaphors, Personal Carbon Allowances, and Consumption Rationing in Historical Perspective." *Climatic Change* 104:199–222.

Entman, R. M. 1993. "Framing: Toward Clarification of a Fractured Paradigm." *Journal of Communication* 43 (4): 51–58.

Ereaut, G., and N. Segnit. 2006. *Warm Words: How are We Telling the Climate Story and Can We Tell It Better*. London: Institute for Public Policy Research.

Flusberg, S. J., T. Matlock, and P. H. Thibodeau. 2017. "Metaphors for the War (or Race) Against Climate Change." *Environmental Communication* 11 (6): 769–783.

Fu, X., and G. Wang. 2022. "Confrontation, Competition, or Cooperation? The China–US Relations Represented in China Daily's Coverage of Climate Change (2010–2019)." *Critical Arts* 36 (1-2): 95–109.

Gong, X., Y. Liu, and T. Sun. 2020. "Evaluating Climate Change Governance Using the "Polity–Policy–Politics" Framework: A Comparative Study of China and the United States." *Sustainability* 12 (16): 6403.

Hellmueller, L., and C. Mellado. 2016. "Watchdogs in Chile and the United States: Comparing the Networks of Sources and Journalistic Role Performances." *International Journal of Communication* 10:3261–3280.

Hellsten, I., A. J. Porter, and B. Nerlich. 2014. "Imagining the Future at the Global and National Scale: A Comparative Study of British and Dutch Press Coverage of Rio 1992 and Rio 2012." *Environmental Communication* 8 (4): 468–488.

Hoffman, A. J. 2011. "The Culture and Discourse of Climate Skepticism." *Strategic Organization* 9 (1): 77–84.

Koteyko, N., and D. Atanasova. 2017. "Metaphor and the Representation of Scientific Issues: Climate Change in Online and Print Media." In *The Routledge Handbook of Metaphor and Language*, edited by E. Semino and Z. Demjén, 296–308. Oxon: Routledge.

Lakoff, G., and M. Johnson. 2008. *Metaphors We Live by*. Chicago: University of Chicago press.

Lee, C.-C., J. M. Chan, Z. Pan, and C. Y. K. So. 2002. *Global Media Spectacle: News War Over Hong Kong*. Hong Kong: Hong Kong University Press.

Liu, M. 2015. "Scapegoat or Manipulated Victim? Metaphorical Representations of the Sino-Us Currency Dispute in Chinese and American Financial News." *Text & Talk* 35 (3): 337–357.

Liu, M., and J. Huang. 2022. ""Climate Change" Vs. "Global Warming": A Corpus-Assisted Discourse Analysis of Two Popular Terms in the New York Times." *Journal of World Languages* 8 (1): 34–55.

Liu, M., and J. Huang. 2024. "Framing Responsibilities for Climate Change in Chinese and American Newspapers: A Corpus-Assisted Discourse Study." *Journalism* 25 (8): 1792–1811.

Luokkanen, M., S. Huttunen, and M. Hildén. 2014. "Geoengineering, News Media and Metaphors: Framing the Controversial." *Public Understanding of Science* 23 (8): 966–981.

McGee, J. 2014. "The Influence of US Neoliberalism on International Climate Change Policy." In *Climate Innovation: Liberal Capitalism and Climate Change*, edited by N. E. Harrison, and J. Mikler, 193–214. Hampshire: Palgrave.

Milne, M. J., K. Kearins, and S. Walton. 2006. "Creating Adventures in Wonderland: The Journey Metaphor and Environmental Sustainability." *Organization* 13 (6): 801–839.

Musolff, A. 2004. *Metaphor and Political Discourse: Analogical Reasoning in Debates About Europe*. Hampshire: Palgrave.

Nerlich, B., and I. Hellsten. 2014. "The Greenhouse Metaphor and the Footprint Metaphor." *TATuP-Zeitschrift für Technikfolgenabschätzung in Theorie und Praxis* 23 (2): 27–33.

Nerlich, B., and N. Koteyko. 2010. "Carbon Gold Rush and Carbon Cowboys: A New Chapter in Green Mythology?" *Environmental Communication* 4 (1): 37–53.

Painter, J. 2013. *Climate Change in the Media: Reporting Risk and Uncertainty*. London: I. B. Tauris.

Pan, Y., M. Opgenhaffen, and B. Van Gorp. 2019. "Negotiating Climate Change: A Frame Analysis of COP21 in British, American, and Chinese News Media." *Public Understanding of Science* 28 (5): 1–15.

Parks, P. 2020. "Is Climate Change a Crisis–and Who Says So? An Analysis of Climate Characterization in Major US News Media." *Environmental Communication* 14 (1): 82–96.

Pragglejaz Group. 2007. "MIP: A Method for Identifying Metaphorically Used Words in Discourse." *Metaphor and Symbol* 22 (1): 1–39.

Reisigl, M., and R. Wodak. 2016. "The Discourse Historical Approach (DHA)." In *Methods of Critical Discourse Studies*, edited by R. Wodak and M. Meyer, 23–61. London: Sage.

Russill, C., and Z. Nyssa. 2009. "The Tipping Point Trend in Climate Change Communication." *Global Environmental Change* 3 (19): 336–344.

Schmid-Petri, H., S. Adam, I. Schmucki, and T. Häussler. 2015. "A Changing Climate of Skepticism: The Factors Shaping Climate Change Coverage in the US Press." *Public Understanding of Science* 26 (4): 498–513.

Selin, H., and S. D. VanDeveer. 2020. "Climate Change Politics and Policy in the United States: Forward, Reverse and Through the Looking Glass." In *Climate Governance Across the Globe*, edited by R. K. W. Wurzel, M. S. Andersen, and P. Tobin, 123–141. Oxon: Routledge.

Song, Y., Z. Huang, J. P. Schuldt, and Y. C. Yuan. 2021. "National Prisms of a Global Phenomenon: A Comparative Study of Press Coverage of Climate Change in the US, UK and China." *Journalism* 23 (10): 2208–2229.

Stoddart, M. C. J., H. Ramos, K. Foster, and T. Yla-Anttila. 2021. "Competing Crises? Media Coverage and Framing of Climate Change During the Covid-19 Pandemic." *Environmental Communication* 17 (3): 276–292.

Su, Y., and J. Hu. 2021. "How Did the Top Two Greenhouse Gas Emitters Depict Climate Change? A Comparative Analysis of the Chinese and US Media." *Public Understanding of Science* 30 (7): 881–897.

van Dijk, T. A. 1998. *Ideology: A Multidisciplinary Approach*. London: Sage.

Wang, G., C. Cui, and Y. Qu. 2024. "Representations of Dual Carbon Goals in English-Language News Reporting by China's Official Media." In *Handbook on Climate Change and Environmental Governance in China*, edited by X. Zang and X. Zhang, 199–219. Cheltenham: Edward Elgar.

Wang, Y., and H. Habil. 2024. "Metaphor Construction of Environmental Issues in Chinese and American News Media." In *Islamic Finance: New Trends in Law and Regulation*, edited by N. Mansour, and L. Bujosa, 701–711. Cham, Switzerland: Springer.

Woods, R., A. Fernández, and S. Coen. 2012. "The use of Religious Metaphors by UK Newspapers to Describe and Denigrate Climate Change." *Public Understanding of Science* 21 (3): 323–339.

Xie, L. 2015. "The Story of Two Big Chimneys: A Frame Analysis of Climate Change in US and Chinese Newspapers." *Journal of Intercultural Communication Research* 44 (2): 151–177.

Yu, Y. 2019. "Metaphorical Representations of "Leftover Women": Between Traditional Patriarchy and Modern Egalitarianism." *Social Semiotics* 31 (2): 248–265.

Zhang, H., and F. Gai. 2022. "Conceptual Metaphors in Chinese and American News Discourses from the Ecolinguistic Perspective." *Modern Foreign Languages (Bimonthly)* 45 (5): 621–633.